

Esercizi - Fondamenti di Teoria Quantistica dei campi

APPUNTI PIENI DI ERRORI, SVISTE, BUCHI E DUBBI. SICURAMENTE GLI APPUNTI IN LATEX CONTENGONO MENO ERRORI E BUCHI, MA NON GARANTISCO NULLA SU SVISTE E, SOPRATTUTTO, DUBBI.

SCATTERING

USEFULL RELATIONS	3
REGOLE DI FEYNMAN	4
YUKAWA	5
QED	6
TEORIA DI YUKAWA :	
$e^{\pm} e^{\pm} \longrightarrow e^{\pm} e^{\pm}$	9
$e^{+} e^{-} \longrightarrow e^{+} e^{-}$	15
$e^{+} e^{-} \longrightarrow 2\phi$	22
$e^{\pm} \phi \longrightarrow e^{\pm} \phi$	30
QED : MÜLLER SCATTERING	
$e^{\pm} e^{\pm} \longrightarrow e^{\pm} e^{\pm}$	37
BIHABHA SCATTERING	
$e^{+} e^{-} \longrightarrow e^{+} e^{-}$	44
COMPTON SCATTERING	
$e^{\pm} \gamma \longrightarrow e^{\pm} \gamma$	51
(NO NAME) SCATTERING	
$e^{+} e^{-} \longrightarrow 2\gamma$	

RELAZIONI UTILI

SPINORI

$$\begin{cases} \sum_s u_\alpha(\vec{p}) \bar{u}_\beta(\vec{p}) = (\not{p} + m)_{\alpha\beta} \\ \sum_s v_\alpha(\vec{p}) \bar{v}_\beta(\vec{p}) = (\not{p} - m)_{\alpha\beta} \end{cases}$$

VARIABILI DI MANDELSTAM

$$\begin{cases} s = (p_1 + p_2)^2 = (p_3 + p_4)^2 \\ t = (p_1 - p_3)^2 = (p_2 - p_4)^2 \\ u = (p_1 - p_4)^2 = (p_2 - p_3)^2 \end{cases}$$

$\Rightarrow$
ALTE ENERGIE
($m \ll p$)

$$\begin{cases} p_1 p_2 = p_3 p_4 = \frac{s}{2} \\ p_1 p_3 = p_2 p_4 = -\frac{t}{2} \\ p_1 p_4 = p_2 p_3 = -\frac{u}{2} \end{cases}$$

SEZIONE D'URTO

$$\left. \frac{d\sigma}{d\Omega} \right|_{cm} = \frac{1}{64\pi^2 s} \frac{|p_i'|}{|p_i|} |\mathcal{M}|^2$$

MATRICI γ

$$\text{Tr} \{ \gamma^\mu \gamma^\nu \} = 4 \eta^{\mu\nu}$$

$$\text{Tr} \{ \gamma^\alpha \gamma^\mu \gamma^\beta \gamma^\nu \} = 4 \left(\eta^{\alpha\mu} \eta^{\beta\nu} + \eta^{\alpha\nu} \eta^{\beta\mu} - \eta^{\alpha\beta} \eta^{\mu\nu} \right)$$

$$\text{Tr} \{ \gamma_\rho \gamma^\nu \gamma_\sigma \gamma^\mu \gamma_\lambda \gamma_\nu \gamma_\tau \gamma_\mu \} = -32 \eta_{\sigma\tau} \eta_{\lambda\rho}$$

REGOLE DI FEYNMAN

ELENCO REGOLE DI FEYNMAN PER LA TEORIA DI YUKAWA E LA QED.

YUKAWA: (SPAZIO DEGLI IMPULSI)

$$\mathcal{L} = \frac{1}{2} \partial_\mu \phi \partial^\mu \phi - \frac{1}{2} m^2 \phi^2 + \bar{\psi} (\not{\partial} - M) \psi - g \phi \bar{\psi} \psi$$

• AD OGNI LINEA INTERNA:

$$\tilde{\Delta}(p) = \frac{i}{p^2 - m^2 + i\varepsilon} \quad (\phi^+ \phi)$$

$$\tilde{S}_{\alpha\beta}(p) = i (\not{p} - M + i\varepsilon)^{-1}_{\alpha\beta} \quad (\psi \bar{\psi})$$

• AD OGNI VERTICE:

$$-ig(2\pi)^4 \delta_{\alpha\beta} \delta^4(\sum p_i)$$

• DIVIDIAMO PER IL FATTORE DI SIMMETRIA.

QED: (SPAZIO DEGLI IMPULSI)

$$\mathcal{L} = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu} - \frac{1}{2\xi} (\partial A)^2 + \bar{\psi} (i \not{D} - m) \psi$$

$$\mathcal{L}_{\text{INT}} = -e A_\mu \bar{\psi} \gamma^\mu \psi$$

- AD OGNI LINEA INTERNA:

$$\tilde{\Delta}_{\mu\nu}(p) = -\frac{i}{k^2} \eta_{\mu\nu} \quad (\xi = 1)$$

$$\tilde{S}_{\alpha\beta}(p) = i (\not{p} - M + i\varepsilon)^{-1}_{\alpha\beta} \quad (\psi \bar{\psi})$$

- AD OGNI VERTICE:

$$-ie(2\pi)^4 \gamma^\mu_{\alpha\beta} \delta^4(\sum p_i)$$

- DIVIDIAMO PER IL FATTORE DI SIMMETRIA.

AMPIEZZE DI PROBABILITÀ $\mathcal{M}$:

PER IL CALCOLO DI $\mathcal{M}$ BASTA APPLICARE LE REGOLE PER IL CALCOLO DELLA FUNZIONE DI GREEN $\tilde{G}$, MA RICORDANDO:

- AMPUTARE LE GAMBE ESTERNE
- INSERIRE LE POLARIZZAZIONI: ($\ell = \text{LEPTONE}$)

$$\phi \sim 1 \quad (\phi)$$

$$\left[\begin{array}{l} \ell_{in}^- \sim u_\alpha(\vec{p}) \end{array} \right. \quad (\bar{4})$$

$$\left[\begin{array}{l} \ell_{out}^- \sim \bar{u}_\alpha(\vec{p}) \end{array} \right. \quad (4)$$

$$\left[\begin{array}{l} \ell_{in}^+ \sim \bar{v}_\alpha(\vec{p}) \end{array} \right. \quad (4)$$

$$\left[\begin{array}{l} \ell_{out}^+ \sim v_\alpha(\vec{p}) \end{array} \right. \quad (\bar{4})$$

$$\left[\begin{array}{l} \gamma_{in} \sim \epsilon_\mu^*(\lambda) \end{array} \right. \quad (A_\mu)$$

$$\left[\begin{array}{l} \gamma_{out} \sim \epsilon_\mu(\lambda) \end{array} \right. \quad (A_\mu)$$

TEORIA DI YUKAWA

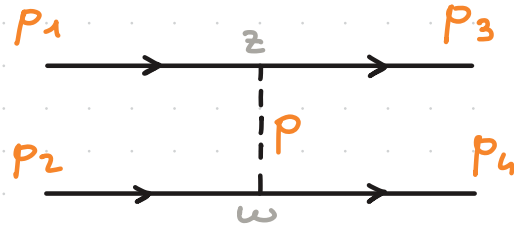
RIFERIMENTI : [SREDNICKI] : CAP. 45 (P. 282 - 290) (REGOLE E SEGNI RELATIVI.)
CAP. 46 + 48 (P. 291 - 293 +
+ P. 298 - 302) (CONTI E RISULTATI)
↓
NEGLI ESERCIZI

YUKAWA :

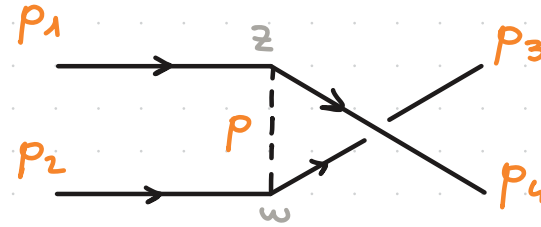
$$e^- \quad e^- \longrightarrow e^- \quad e^-$$

$$\left(\begin{array}{c} \text{FUNZIONA} \\ e^+ \quad e^+ \end{array} \longrightarrow \begin{array}{c} \text{UGUALE} \\ e^+ \quad e^+ \end{array} \text{ PER } \right)$$

I DIAGRAMMI SONO :



CANALE t : $p = p_1 - p_3$



CANALE u : $p = p_1 - p_4$

POSSIAMO VEDERE IL SEGNO DEI DUE DIAGRAMMI CON LE CONTRAZIONI :

$$t: \quad \langle 0 | T \left[\underbrace{\phi(x_1) \phi(x_2)}_{\text{contraction}} \underbrace{\phi(x_3) \phi(x_4)}_{\text{contraction}} \underbrace{\phi(z) \phi(z)}_{\text{contraction}} \underbrace{\phi(w) \phi(w)}_{\text{contraction}} \right] | 0 \rangle$$

$\phi(z) \phi(x_1)$	5 SCAMBI
$\phi(w) \phi(x_2)$	5 SCAMBI
$\phi(x_3) \phi(z)$	1 SCAMBI
$\phi(x_4) \phi(w)$	0 SCAMBI

$$\Rightarrow (-1)^{11} = -1$$

$$\mu: \langle 0 | T [\overline{\psi}(x_1) \overline{\psi}(x_2) \psi(x_3) \psi(x_4) \phi(z) \overline{\psi}(z) \psi(z) \phi(w) \overline{\psi}(w) \psi(w)] | 0 \rangle$$

$$\begin{cases} \psi(z) \overline{\psi}(x_1) & 5 \text{ SCAMBI} \\ \psi(w) \overline{\psi}(x_2) & 5 \text{ SCAMBI} \\ \psi(x_3) \overline{\psi}(w) & 2 \text{ SCAMBI} \\ \psi(x_4) \overline{\psi}(z) & 0 \text{ SCAMBI} \end{cases}$$

$$\Rightarrow (-1)^{12} = +1$$

DUNQUE C'È UN SEGNO DI DIFFERENZA E DOVREMO CALCOLARE:

$$|\mathcal{M}|^2 = \frac{1}{4} \sum_s \left\{ |\mathcal{M}_t|^2 + |\mathcal{M}_\mu|^2 - 2 \operatorname{Re} \mathcal{M}_t^* \mathcal{M}_\mu \right\}$$

IN CU, ABBIAMO:

$$\mathcal{M}_t = \frac{i(-ig)^2}{(p_1 - p_3)^2 - m^2} \left(\bar{u}_\alpha(p_1) \delta_{\alpha\beta} u_\beta(p_3) \right) \left(\bar{u}_\gamma(p_2) \delta_{\gamma\lambda} u_\lambda(p_4) \right)$$

$$\mathcal{M}_\mu = \frac{i(-ig)^2}{(p_1 - p_4)^2 - m^2} \left(\bar{u}_\alpha(p_1) \delta_{\alpha\beta} u_\beta(p_4) \right) \left(\bar{u}_\gamma(p_2) \delta_{\gamma\lambda} u_\lambda(p_3) \right)$$

CALCULANDO

$$\begin{aligned}
 \frac{1}{4} \sum_s |\mathcal{M}_t|^2 &= \frac{(-ig)^4}{4(t-m^2)^2} \sum_s (\bar{u}_\alpha(p_1) u_\alpha(p_3)) (\bar{u}_\delta(p_2) u_\delta(p_4)) \times \\
 &\quad \times (\bar{u}_\beta(p_3) u_\beta(p_1)) (\bar{u}_\delta(p_4) u_\delta(p_2)) \\
 &= \frac{(-ig)^4}{4(t-m^2)^2} (\not{p}_3 - m)_{\alpha\beta} (\not{p}_1 - m)_{\beta\alpha} (\not{p}_2 - m)_{\delta\gamma} (\not{p}_4 - m)_{\gamma\delta} \\
 &= \frac{(-ig)^4}{4t^2} \text{Tr}(\not{p}_3 \not{p}_1) \text{Tr}(\not{p}_2 \not{p}_4) \\
 &= \frac{(-ig)^4}{4t^2} 16(p_3 p_1)(p_2 p_4) \\
 &= + \frac{g^4}{4t^2} 16 \frac{t^2}{4} \\
 &= + g^4
 \end{aligned}$$

$$\frac{1}{4} \sum |\mathcal{M}_\mu|^2 = \frac{(-ig)^4}{4(\mu - m^2)^2} \sum_5 (\bar{\mu}_\alpha(p_1) \mu_\alpha(p_4)) (\bar{\mu}_\delta(p_2) \mu_\delta(p_3)) \times$$

$$\times (\bar{\mu}_\beta(p_4) \mu_\beta(p_1)) (\bar{\mu}_\delta(p_3) \mu_\delta(p_2))$$

$$= \frac{(-ig)^4}{4(\mu - m^2)^2} (\not{p}_4 - m)_{\alpha\beta} (\not{p}_1 - m)_{\beta\alpha} (\not{p}_2 - m)_{\delta\gamma} (\not{p}_3 - m)_{\gamma\delta}$$

$$= \frac{(-ig)^4}{4\mu^2} \text{Tr}(\not{p}_4 \not{p}_1) \text{Tr}(\not{p}_2 \not{p}_3)$$

$$= \frac{(-ig)^4}{\mu^2} 16(p_4 p_1)(p_2 p_3)$$

$$= + \frac{g^4}{4\mu^2} 16 \frac{\mu^2}{4}$$

$$= + g^4$$

ORA

$$\begin{aligned}
 \frac{1}{4} \int_S \mathcal{M}_t^* \mathcal{M}_\mu &= \frac{1}{4} (-ig)^4 \int_S (\bar{u}_\alpha(p_3) u_\alpha(p_1)) (\bar{u}_\beta(p_4) u_\beta(p_2)) \times \\
 &\times \frac{+1}{[t-m^2][\mu-m^2]} \times (\bar{u}_\gamma(p_1) u_\gamma(p_4)) (\bar{u}_\delta(p_2) u_\delta(p_3)) \\
 &= + \frac{g^4}{4t\mu} (\not{p}_1 - m)_{\alpha\gamma} (\not{p}_4 - m)_{\gamma\beta} (\not{p}_2 - m)_{\beta\delta} (\not{p}_3 - m)_{\delta\alpha} \\
 &\simeq + \frac{g^4}{4t\mu} p_{1\mu} p_{4\nu} p_{2\rho} p_{3\sigma} \text{Tr} \{ \gamma^\mu \gamma^\nu \gamma^\rho \gamma^\sigma \} \\
 &= + \frac{g^4}{4t\mu} p_{1\mu} p_{4\nu} p_{2\rho} p_{3\sigma} 4 (\eta^{\mu\nu} \eta^{\rho\sigma} + \eta^{\mu\sigma} \eta^{\nu\rho} - \eta^{\mu\rho} \eta^{\nu\sigma}) \\
 &= + \frac{g^4}{t\mu} \{ (p_1 p_4)(p_2 p_3) + (p_1 p_3)(p_2 p_4) - (p_1 p_2)(p_3 p_4) \} \\
 &= + \frac{g^4}{t\mu} \left\{ \frac{\mu^2}{4} + \frac{t^2}{4} - \frac{s^2}{4} \right\} \\
 &= + \frac{g^4}{4t\mu} (\mu^2 + t^2 - (t+\mu)^2)
 \end{aligned}$$

$$= + \frac{g^4}{4tu} (-2tu)$$

$$= -\frac{g^4}{2}$$

DUNQUE

$$|\mathcal{M}|^2 = g^4 \left\{ 1 + 1 + 2 \cdot \frac{1}{2} \right\} = 3g^4$$

RISULTATO SREDNICKI:

$$|\mathcal{M}|^2 \sim 3g^4 \quad \checkmark$$

(CAMBIA L'ESPRESSIONE RISPETTO $e^+e^- \rightarrow e^+e^-$
SE NON GUARDIAMO IL LIMITE AD ALTE
ENERGIE)

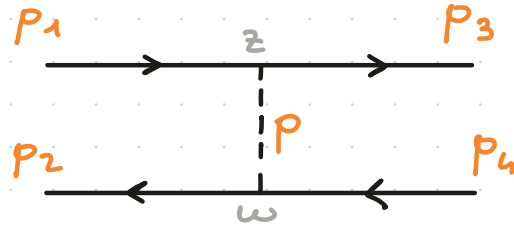
LA SEZIONE D'URTO:

$$\left. \frac{d\sigma}{d\Omega} \right|_{\text{cm}} = \frac{|\mathcal{M}|^2}{64\pi^2 (p_1 + p_2)^2} = \frac{3g^4}{64\pi^2 s}$$

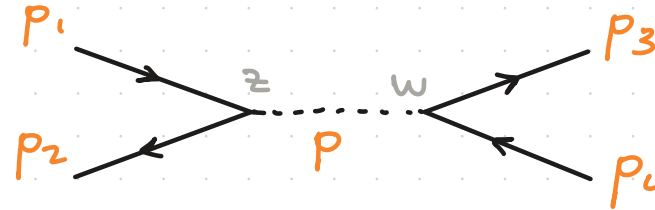
YUKAWA :

$$e^+ e^- \longrightarrow e^+ e^-$$

I DIAGRAMMI SONO :



CANALE t: $p = p_1 - p_3$



CANALE s: $p = p_1 + p_2$

POSSIAMO VEDERE IL SEGNO DEI DUE DIAGRAMMI CON LE CONTRAZIONI :

$$t: \langle 0 | T [\underbrace{\phi(x_1) \phi(x_2)}_{\text{contraction}}, \underbrace{\phi(x_3) \phi(x_4)}_{\text{contraction}}, \underbrace{\phi(z) \phi(w)}_{\text{contraction}}, \underbrace{\phi(z) \phi(w)}_{\text{contraction}}] | 0 \rangle$$

$\phi(z) \phi(x_1)$	5 SCAMBI
$\phi(x_2) \phi(w)$	3 SCAMBI
$\phi(x_3) \phi(z)$	1 SCAMBI
$\phi(w) \phi(x_4)$	0 SCAMBI

$$\Rightarrow (-1)^9 = -1$$

$$s. \quad \langle 0 | T \left[\overline{\psi}(x_1) \psi(x_2) \psi(x_3) \overline{\psi}(x_4) \overbrace{\phi(z) \overline{\psi}(z) \psi(z) \phi(w) \overline{\psi}(w) \psi(w)} \right] | 0 \rangle$$

$$\begin{cases} \psi(z) \overline{\psi}(x_1) & 5 \text{ SCAMB} \\ \psi(x_2) \overline{\psi}(z) & 2 \text{ SCAMB} \\ \psi(x_3) \overline{\psi}(w) & 1 \text{ SCAMB} \\ \psi(w) \overline{\psi}(x_4) & 0 \text{ SCAMB} \end{cases} \Rightarrow (-1)^8 = +1$$

CALCOLARE QUINDI

$$|\mathcal{M}|^2 = \frac{1}{4} \sum_s \left\{ |\mathcal{M}_s|^2 + |\mathcal{M}_t|^2 - 2 \operatorname{Re} \mathcal{M}_s^* \mathcal{M}_t \right\}$$

con

$$\mathcal{M}_t = \frac{(-ig)^2 i}{(p_1 - p_3)^2 - m^2} \left(\overline{u}_\alpha(p_3) u_\alpha(p_1) \right) \left(\overline{v}_\beta(p_2) v_\beta(p_4) \right)$$

$$\mathcal{M}_s = \frac{(-ig)^2 i}{(p_1 + p_2)^2 - m^2} \left(\overline{v}_\gamma(p_2) u_\gamma(p_1) \right) \left(\overline{u}_\delta(p_3) v_\delta(p_4) \right)$$

CALCOLIAMO

$$\frac{1}{4} \sum_s |\mathcal{M}_t|^2 = \frac{g^4}{4(t-m^2)^2} \sum_s \left(\bar{u}_\alpha(p_1) u_\alpha(p_3) \right) \left(\bar{v}_\beta(p_4) v_\beta(p_2) \right) \times$$

$$\times \left(\bar{u}_\gamma(p_3) u_\gamma(p_1) \right) \left(\bar{v}_\delta(p_2) v_\delta(p_4) \right)$$

$$= \frac{g^4}{4(t-m^2)^2} (\not{p}_3 - m)_{\alpha\gamma} (\not{p}_1 - m)_{\gamma\alpha} \times$$

$$\times (\not{p}_2 + m)_{\beta\delta} (\not{p}_4 + m)_{\delta\beta}$$

$$\sim \frac{g^4}{4t^2} \text{Tr}\{\not{p}_3 \not{p}_1\} \text{Tr}\{\not{p}_2 \not{p}_4\}$$

$$= \frac{g^4}{4t^2} 16(p_1 p_3)(p_2 p_4)$$

$$= \frac{4g^4}{t^2} \cdot \frac{t^2}{4}$$

$$= g^4$$

$$\frac{1}{4} \frac{1}{s} |\mathcal{M}_s|^2 = \frac{g^4}{4(s-m^2)^2} \frac{1}{s} \left(\bar{u}_\alpha(p_1) v_\alpha(p_2) \right) \left(\bar{v}_\beta(p_4) u_\beta(p_3) \right) \times \\ \times \left(\bar{v}_\gamma(p_2) u_\gamma(p_1) \right) \left(\bar{u}_\delta(p_3) v_\delta(p_4) \right)$$

$$= \frac{g^4}{4(s-m^2)^2} (\not{p}_2 + m)_{\alpha\gamma} (\not{p}_1 - m)_{\gamma\alpha} \times \\ (\not{p}_3 - m)_{\beta\delta} (\not{p}_4 + m)_{\delta\beta}$$

$$\sim \frac{g^4}{4s^2} \text{Tr}\{\not{p}_2 \not{p}_1\} \text{Tr}\{\not{p}_3 \not{p}_4\}$$

$$= \frac{g^4}{4s^2} 16 (p_1 p_2) (p_3 p_4)$$

$$= \frac{4g^4}{s^2} \frac{s^2}{4}$$

$$= g^4$$

$$\frac{1}{4} \sum \mathcal{M}_s^+ \mathcal{M}_t = \frac{(-i)^4 g^4}{4[s-m^2][t-m^2]} \sum_s \left(\bar{u}_\gamma(p_1) v_\gamma(p_2) \right) \left(\bar{v}_\delta(p_4) u_\delta(p_3) \right) \times \\ \times \left(\bar{u}_\alpha(p_3) u_\alpha(p_1) \right) \left(\bar{v}_\beta(p_2) v_\beta(p_4) \right)$$

$$= \frac{+g^4}{4[s-m^2][t-m^2]} (\not{p}_2 + m)_{\gamma\beta} (\not{p}_4 + m)_{\beta\delta} (\not{p}_3 - m)_{\delta\alpha} (\not{p}_1 - m)_{\alpha\gamma}$$

$$\sim \frac{+g^4}{4st} \text{Tr} \{ \not{p}_2 \not{p}_4 \not{p}_3 \not{p}_1 \}$$

$$= \frac{+g^4}{4st} p_{2\mu} p_{4\nu} p_{3\rho} p_{1\sigma} 4(\eta^{\mu\nu} \eta^{\rho\sigma} + \eta^{\mu\sigma} \eta^{\nu\rho} - \eta^{\mu\rho} \eta^{\nu\sigma})$$

$$= \frac{+g^4}{st} \left\{ (p_2 p_4)(p_1 p_3) + (p_1 p_2)(p_3 p_4) - (p_2 p_3)(p_1 p_4) \right\}$$

$$= \frac{+g^4}{st} \left(\frac{t^2}{4} + \frac{s^2}{4} - \frac{\mu^2}{4} \right)$$

$$= \frac{+g^4}{st4} (t^2 + s^2 - (s+t)^2)$$

$$= -\frac{g^4}{2}$$

DUNQUE

$$|\mathcal{M}|^2 = g^4 + g^4 + 2 \cdot \frac{g^4}{2} = 3g^4$$

RISULTATO SREDNICKI:

$$|\mathcal{M}|^2 = g^4 \left\{ \frac{(s-4m^2)^2}{(\mu^2-s)^2} + \frac{st-4m^2\mu}{(\mu^2-s)(\mu^2-t)} + \frac{(t-4m^2)^2}{(\mu^2-t)^2} \right\} \sim g^4 \{1+1+1\} \checkmark$$

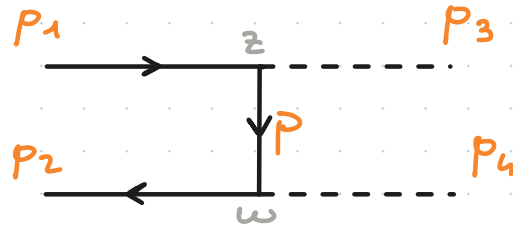
LA SEZIONE D'URTO È :

$$\left. \frac{d\sigma}{d\Omega} \right|_{cm} = \frac{|M|^2}{64\pi^2 s} = \frac{3g^4}{64\pi^2 s}$$

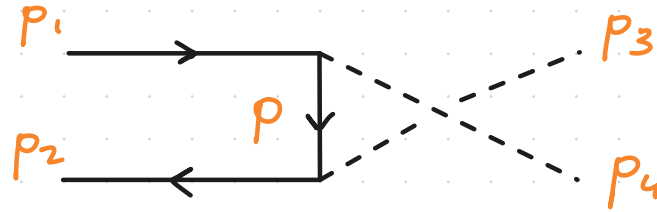
YUKAWA :

$$e^+ e^- \longrightarrow 2\phi$$

I DIAGRAMMI SONO :



CANALE t : $p = p_1 - p_3$



CANALE u : $p = p_1 - p_4$

POSSIAMO VEDERE IL SEGNO DEI DUE DIAGRAMMI CON LE CONTRAZIONI :

$$t: \langle 0 | T [\bar{\psi}(x_1) \psi(x_2) \phi(x_3) \phi(x_4) \phi(z) \bar{\psi}(z) \psi(z) \phi(w) \bar{\psi}(w) \psi(w)] | 0 \rangle$$

$\psi(z) \bar{\psi}(x_1)$	3 SCAMBI
$\psi(x_2) \bar{\psi}(w)$	1 SCAMBI
$\psi(w) \bar{\psi}(z)$	0 SCAMBI

$$\Rightarrow (-1)^4 = +1$$

$$\mu: \langle 0 | T \left[\underbrace{\overline{\psi}(x_1) \psi(x_2)}_{\text{1 swap}} \underbrace{\overline{\psi}(x_3) \psi(x_4)}_{\text{1 swap}} \underbrace{\overline{\psi}(z) \psi(z)}_{\text{1 swap}} \underbrace{\overline{\psi}(\omega) \psi(\omega)}_{\text{1 swap}} \right] | 0 \rangle$$

$$\begin{cases} \overline{\psi}(z) \psi(x_1) & 3 \text{ swaps} \\ \overline{\psi}(x_2) \psi(\omega) & 1 \text{ swap} \\ \overline{\psi}(\omega) \psi(z) & 0 \text{ swaps} \end{cases} \Rightarrow (-1)^4 = +1$$

QUINDI NON C'È SEGNO RELATIVO E DOBBIAMO CALCOLARE:

$$|\mathcal{M}|^2 = \frac{1}{4} \frac{1}{s} \left\{ |\mathcal{M}_t|^2 + |\mathcal{M}_\mu|^2 + 2 \operatorname{Re} \mathcal{M}_t^\dagger \mathcal{M}_\mu \right\}$$

con

$$\mathcal{M}_t = (-ig)^2 i \left(\bar{v}_\alpha(p_2) (\not{p}_t - m + i\varepsilon)^{-1}_{\alpha\beta} u_\beta(p_1) \right)$$

$$\mathcal{M}_\mu = (-ig)^2 i \left(\bar{v}_\alpha(p_2) (\not{p}_\mu - m + i\varepsilon)^{-1}_{\alpha\beta} u_\beta(p_1) \right)$$

SAPENDO $i(\not{p} - m)^{-1}_{\alpha\beta} = \frac{i(\not{p} + m)_{\alpha\beta}}{p^2 - m^2 + i\varepsilon}$

DUNQUE

$$\frac{1}{4} \sum_s |\mathcal{M}_t|^2 = \frac{(-ig)^4}{4(p_t^2 - m^2)^2} \sum_s \left(\bar{v}_\alpha(p_2) (\not{p}_t + m)_{\alpha\beta} u_\beta(p_1) \right) \times$$

$$\times \left(\bar{u}_\delta(p_1) (\not{p}_t + m)_{\delta\gamma} v_\gamma(p_2) \right)$$

$$\sim \frac{g^4}{4t^2} \text{Tr} \{ \not{p}_2 \not{p}_t \not{p}_1 \not{p}_t \}$$

$$\sim \frac{g^4}{4t^2} 4 \left\{ 2 [p_2(p_1 - p_3)] [p_1(p_1 - p_3)] + \right.$$

$$\left. - (p_1 - p_3)(p_1 - p_3)(p_2 p_1) \right\}$$

$$= \frac{g^4}{t^2} \left\{ 2 [p_2(p_1 - p_3)] [p_1^2 - p_1 p_3] - t(p_1 p_2) \right\}$$

$$p_1^2 \sim 0 \quad \sim \frac{g^4}{t^2} \left\{ -2(p_1 p_3) [p_1 p_2 - p_2 p_3] - \frac{5t}{2} \right\}$$

$$= \frac{g^4}{t^2} \left\{ + 2 \frac{t}{2} \left[\frac{s}{2} + \frac{\mu}{2} \right] - \frac{st}{2} \right\}$$

$$= \frac{g^4}{t^2} \left\{ \frac{t}{2} (s + \mu) - \frac{st}{2} \right\}$$

$$= \frac{g^4}{t^2} \frac{1}{2} (-t^2 - st)$$

$$= -\frac{g^4}{2t} (t + s)$$

$$= + \frac{g^4}{2} \frac{\mu}{t}$$

IL CONTO DI $|\mathcal{M}_\mu|^2$ È IDENTICO MA CON IL CAMBIO
 $p_t \rightarrow p_\mu$

DUNQUE

$$\frac{1}{4} \frac{\Gamma}{s} |\mathcal{M}_\mu|^2 = \frac{(-ig)^4}{4(\mu - m^2 + i\varepsilon)^2} \frac{\Gamma}{s} \left(\bar{v}_\alpha(p_2) (\not{p}_\mu - m)_{\alpha\beta} u_\beta(p_1) \right) \times \\ \times \left(\bar{u}_\delta(p_1) (\not{p}_\mu - m)_{\delta\gamma} v_\gamma(p_2) \right)$$

$$\sim \frac{g^4}{4\mu^2} \text{Tr} \{ \not{p}_1 \not{p}_\mu \not{p}_2 \not{p}_\mu \}$$

$$= \frac{g^4}{4\mu^2} 4 \left[(p_1 p_\mu) (p_2 p_\mu) + (p_1 p_\mu) (p_2 p_\mu) - p_\mu^2 (p_1 p_2) \right]$$

$$= \frac{g^4}{\mu^2} \left[2 p_1 (p_1 - p_4) \cdot p_2 (p_1 - p_4) - \mu \frac{s}{2} \right]$$

$$= \frac{g^4}{\mu^2} \left[-2 (p_1 p_4) [p_1 p_2 - p_2 p_4] - \frac{\mu s}{2} \right]$$

$$= \frac{g^4}{\mu^2} \left[+\mu \left[\frac{s}{2} + \frac{t}{2} \right] - \frac{\mu s}{2} \right]$$

$$= \frac{g^4}{2\mu} (-\mu - s) = \frac{g^4}{2\mu} t$$

ORA

$$\frac{1}{4} \frac{\sum}{s} \mathcal{M}_t^+ \mathcal{M}_\mu = \frac{+(-ig)^4}{4[t-m^2][\mu-m^2]} \frac{\sum}{s} \left(\bar{u}_\alpha(p_1) (\not{p}_t + m)_{\alpha\beta} v_\beta(p_2) \right) \times$$

$$\times \left(\bar{v}_\gamma(p_2) (\not{p}_\mu + m)_{\gamma\delta} u_\delta(p_1) \right)$$

$$\sim \frac{+g^4}{4 t \mu} (\not{p}_2)_{\beta\gamma} p_{\mu\nu} \gamma_{\gamma\delta}^\mu (\not{p}_1)_{\delta\alpha} p_{t\nu} \gamma_{\alpha\beta}^\nu$$

$$= \frac{+g^4}{4 t \mu} p_{\mu\mu} p_{t\nu} p_{2\gamma} p_{1\sigma} \text{Tr} \{ \gamma^\beta \gamma^\mu \gamma^\sigma \gamma^\nu \}$$

$$= + \frac{g^4}{t \mu} p_{\mu\mu} p_{t\nu} p_{2\gamma} p_{1\sigma} (\eta^{\beta\mu} \eta^{\sigma\nu} + \eta^{\beta\nu} \eta^{\mu\sigma} - \eta^{\beta\sigma} \eta^{\mu\nu})$$

$$= + \frac{g^4}{t \mu} \left\{ (p_\mu p_2)(p_t p_1) + (p_t p_2)(p_\mu p_1) - (p_1 p_2)(p_t p_\mu) \right\}$$

$$= \frac{g^4}{t \mu} \left\{ [p_2(p_1 - p_4)][p_1(p_1 - p_3)] + [p_2(p_1 - p_3)][p_1(p_1 - p_4)] - \right.$$

$$\left. - \frac{s}{2} (p_1 - p_3)(p_1 - p_4) \right\}$$

$$= \frac{g^4}{tm} \left\{ -(p_1 p_3) [p_1 p_2 - p_2 p_4] - (p_1 p_4) [p_1 p_2 - p_2 p_3] - \frac{s}{2} [-p_1 p_4 - p_1 p_3 + p_3 p_4] \right\}$$

$$= \frac{g^4}{tm} \left\{ \frac{t}{2} \left[\frac{s}{2} + \frac{t}{2} \right] + \frac{\mu}{2} \left[\frac{s}{2} + \frac{\mu}{2} \right] - \frac{s}{2} \underbrace{\left[\frac{\mu}{2} + \frac{t}{2} + \frac{s}{2} \right]}_{\sim 0} \right\}$$

$$= \frac{+g^4}{4tm} \left\{ t(-\mu) + \mu(-t) \right\}$$

$$= -\frac{g^4}{2}$$

VEDIAMO QUINDI

$$|\mathcal{M}|^2 = g^4 \left(\frac{1}{2} \frac{t}{\mu} + \frac{1}{2} \frac{\mu}{t} - 1 \right)$$

$$= \frac{g^4}{2} \left(\frac{t}{\mu} + \frac{\mu}{t} \right) - g^4$$

RISULTATO SREDNICKI:

$$|\mathcal{M}|^2 = \frac{g^4}{2} \left(\frac{t}{\mu} + \frac{\mu}{t} \right) - g^4$$



QUINDI LA SEZIONE D'URTO DIFFERENZIALE

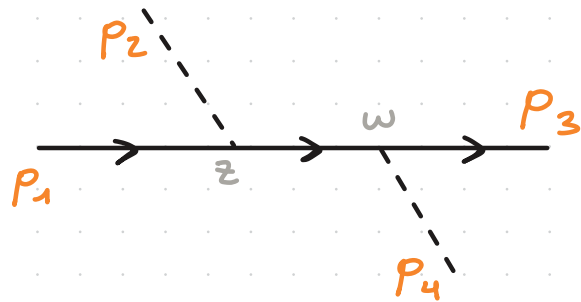
$$\left. \frac{d\sigma}{d\Omega} \right|_{cm} = \frac{g^4}{64\pi S} \left\{ \frac{1}{2} \left(\frac{t}{u} + \frac{u}{t} \right) - 1 \right\}$$

YUKAWA :

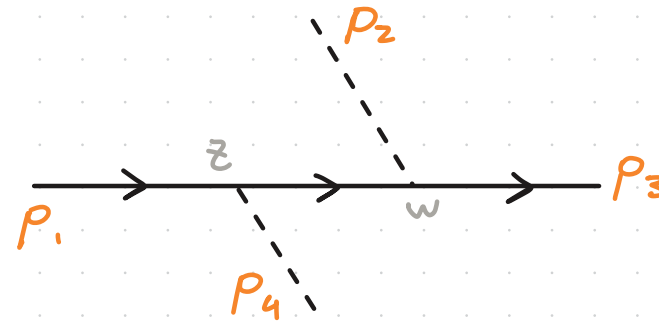
$$e^- \phi \longrightarrow e^- \phi$$

(FUNZIONA UGUALE
PER $e^+ \phi \rightarrow e^+ \phi$)

I DIAGRAMMI SONO :



CANALE S: $p = p_1 + p_2$



CANALE u: $p = p_1 - p_4$

CONTROLLIAMO IL SEGNO DEI DIAGRAMMI :

$$S : \langle 0 | T [\bar{\psi}(x_1) \phi(x_2) \bar{\psi}(x_3) \phi(x_4) \phi(z) \bar{\psi}(z) \bar{\psi}(z) \phi(\omega) \bar{\psi}(\omega) \bar{\psi}(\omega)] | 0 \rangle$$

$$\begin{cases} \bar{\psi}(z) \bar{\psi}(x_1) & 3 \text{ SCAMBI} \\ \bar{\psi}(x_4) \bar{\psi}(\omega) & 1 \text{ SCAMBI} \\ \bar{\psi}(\omega) \bar{\psi}(z) & 0 \text{ SCAMBI} \end{cases}$$

$$\Rightarrow (-1)^4 = +1$$

$$S : \langle 0 | T \left[\bar{\psi}(x_1) \psi(x_2) \bar{\psi}(x_3) \psi(x_4) \bar{\psi}(z) \psi(z) \bar{\psi}(w) \psi(w) \right] | 0 \rangle$$

$$\begin{cases} \bar{\psi}(z) \psi(x_1) & 3 \text{ SCAMBI} \\ \bar{\psi}(x_3) \psi(w) & 1 \text{ SCAMBI} \\ \bar{\psi}(w) \psi(z) & 0 \text{ SCAMBI} \end{cases} \Rightarrow (-1)^4 = +1$$

DOBBIAMO DUNQUE CALCOLARE :

$$|M|^2 = \frac{1}{2} \left\{ |M_s|^2 + |M_u|^2 + 2 \operatorname{Re} M_s^\dagger M_u \right\}$$

CON

$$M_s = \frac{(-ig)^2 i}{p_s^2 - m^2 + i\epsilon} \left(\bar{u}(p_3) (\not{p}_s + m) u(p_1) \right)$$

$$M_u = \frac{(-ig)^2 i}{p_u^2 - m^2 + i\epsilon} \left(\bar{u}(p_3) (\not{p}_u + m) u(p_1) \right)$$

POSSIAMO QUINDI CALCOLARE

$$\frac{1}{2} \int_s |\mathcal{M}_s|^2 = \frac{(-ig)^4}{2(s-m^2)^2} \int_s \left(\bar{u}(p_1) (\not{p}_s + m) u(p_3) \right) \times \\ \times \left(\bar{u}(p_3) (\not{p}_s + m) u(p_1) \right)$$

$$\sim \frac{g^4}{2s^2} \text{Tr} \{ \not{p}_3 \not{p}_s \not{p}_1 \not{p}_s \}$$

$$= \frac{g^4}{2s^2} 4 \left\{ (p_3 p_s)(p_1 p_s) + (p_3 p_s)(p_1 p_s) - (p_1 p_3) p_s^2 \right\}$$

$$\begin{aligned} p_s &= p_1 + p_2 \\ &= p_3 + p_4 \end{aligned} \sim 2 \frac{g^4}{s^2} \left\{ 2[p_3(p_3 + p_4)][p_1(p_1 + p_2)] + \frac{ts}{2} \right\}$$

$$= 2 \frac{g^4}{s^2} \left\{ 2(p_3 p_4)(p_1 p_2) + \frac{ts}{2} \right\}$$

$$= 2 \frac{g^4}{s^2} \left\{ \frac{s^2}{2} + \frac{ts}{2} \right\}$$

$$= \frac{g^4}{s} (s + t)$$

$$= -g^4 \frac{\mu}{s}$$

$$\begin{aligned}
\frac{1}{2} \int \frac{d^4 p}{s} |\mathcal{M}_\mu|^2 &= \frac{(-ig)^4}{2(p_\mu^2 - m^2 + i\epsilon)^2} \int \frac{d^4 p}{s} \left(\bar{u}(p_3) (\not{p}_\mu + m) u(p_1) \right) \left(\bar{u}(p_1) (\not{p}_\mu + m) u(p_3) \right) \\
&\sim \frac{g^4}{2\mu^2} \text{Tr} \{ \not{p}_1 \not{p}_\mu \not{p}_3 \not{p}_\mu \} \\
&= \frac{g^4}{2\mu^2} 4 \left\{ (p_1 p_\mu)(p_3 p_\mu) + (p_1 p_\mu)(p_3 p_\mu) - (p_1 p_3) p_\mu^2 \right\} \\
&= 2 \frac{g^4}{\mu^2} \left\{ 2 [p_1(p_1 - p_4)] [p_3(p_3 - p_2)] + \frac{t}{2} \mu \right\} \\
&= 2 \frac{g^4}{\mu^2} \left\{ + 2(p_1 p_4)(p_2 p_3) + \frac{t\mu}{2} \right\} \\
&= 2 \frac{g^4}{\mu^2} \left\{ + \frac{\mu^2}{2} + \frac{t\mu}{2} \right\} \\
&= -g^4 \frac{s}{\mu}
\end{aligned}$$

$$\frac{1}{2} \Gamma M_s^\dagger M_\mu = \frac{(-i) i (-ig)^4}{2[p_\mu^2 - m^2][p_s^2 - m^2]} \Gamma \left(\bar{u}(p_1) (\not{p}_s + m) u(p_3) \right) \times$$

$$\times \left(\bar{u}(p_3) (\not{p}_\mu + m) u(p_1) \right)$$

$$\sim \frac{+g^4}{2ms} \text{Tr} \{ \not{p}_3 \not{p}_\mu \not{p}_1 \not{p}_s \}$$

$$= +\frac{g^4}{2ms} 4 \{ (p_3 p_\mu)(p_1 p_s) + (p_3 p_s)(p_1 p_\mu) - (p_1 p_3)(p_\mu p_s) \}$$

$$= +\frac{2g^4}{ms} \left\{ -(p_2 p_3)(p_1 p_2) + [p_3(p_3 + p_4)][p_1(p_1 - p_4)] - \right.$$

$$\left. -(p_1 p_3)[(p_1 - p_4)(p_1 + p_2)] \right\}$$

$$= +\frac{2g^4}{ms} \left\{ \frac{s\mu}{4} - (p_3 p_4)(p_1 p_4) - (p_1 p_3)[(p_1 p_2) - (p_4 p_1) - (p_2 p_4)] \right\}$$

$$= +\frac{2g^4}{ms} \left\{ \frac{s\mu}{4} + \frac{s\mu}{4} + \frac{t}{2} \left[\frac{s}{2} + \frac{\mu}{2} + \frac{t}{2} \right] \right\}$$

$$= +g^4$$

DUNQUE

$$|\mathcal{M}|^2 = -g^4 \left(\frac{\mu}{s} + \frac{s}{\mu} \right) + 2g^4$$

RISULTATO

SREDNICKI :

$$\begin{aligned} |\mathcal{M}|^2 &= g^4 \left\{ \frac{1}{(m^2 - s)^2} \left[-s\mu + m^2(gs + \mu) + 7m^4 - 8m^2\mathcal{H}^2 + \mathcal{H}^4 \right] + \right. \\ &\quad + \frac{2}{(m^2 - s)(\mathcal{H}^2 - \mu)} \left[s\mu + 3m^2(s + \mu) + 9m^4 - 8m^2\mathcal{H}^2 - \mathcal{H}^4 \right] + \\ &\quad \left. + \frac{1}{(m^2 - \mu)^2} \left[-s\mu + m^2(9\mu + s) + 7m^4 - 8m^2\mathcal{H}^2 + \mathcal{H}^4 \right] \right\} \\ &\sim g^4 \left\{ \frac{1}{s^2} (-s\mu) + \frac{2}{s\mu} (s\mu) + \frac{1}{\mu^2} (-s\mu) \right\} \\ &= -g^4 \left(\frac{\mu}{s} + \frac{s}{\mu} \right) + 2g^4 \end{aligned}$$


È LA SEZIONE D'URTO DIFFERENZIALE

$$\left. \frac{d\sigma}{d\Omega} \right|_{cm} = \frac{g^4}{64\pi s} \left\{ -\left(\frac{\mu}{s} + \frac{s}{\mu} \right) + 2 \right\}$$

QED

RIFERIMENTI : [SREDNICKI] : CAP. 58 (P. 354 - 356)
CAP. 59 (P. 357 - 361)
CAP. 60 (P. 362 - 370)

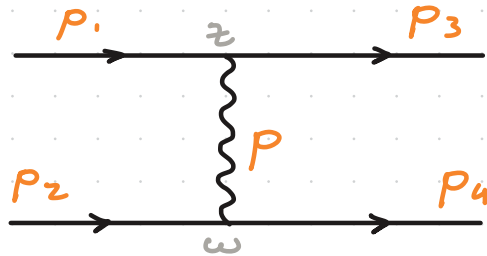
(REGOLE)
(RISULTATI)  NEGLI ESERCIZI
(SEGNI RELATIVI DIAGRAMMI E
RISULTATI)

MULLER SCATTERING

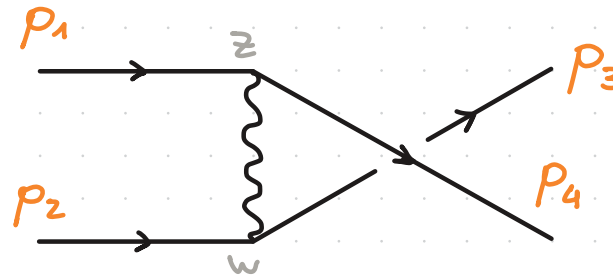
$$e^- e^- \longrightarrow e^- e^-$$

(FUNZIONA UGUALE
PER $e^+ e^+ \rightarrow e^+ e^+$)

I DIAGRAMMI SONO:



CANALE t : $p = p_1 - p_3$



CANALE u : $p = p_1 - p_4$

CONTROLLIAMO IL SEGNO DEI DIAGRAMMI:

$$t. \quad \langle 0 | T \left[\underbrace{\psi(x_1) \psi(x_2)}_{\text{1 swap}} \underbrace{\psi(x_3) \psi(x_4)}_{\text{1 swap}} \underbrace{A_\mu(z) \psi(z)}_{\text{5 swaps}} \underbrace{\psi(z) A_\mu(\omega)}_{\text{5 swaps}} \psi(\omega) \psi(\omega) \right] | 0 \rangle$$

{	$\psi(z) \psi(x_1)$	5	SCAMBI
	$\psi(\omega) \psi(x_2)$	5	SCAMBI
	$\psi(x_3) \psi(z)$	1	SCAMBI
	$\psi(x_4) \psi(\omega)$	0	SCAMBI

$$\Rightarrow (-1)^{11} = -1$$

$$\mu. \quad \langle 0 | T \left[\underbrace{\underbrace{\underbrace{\underbrace{\psi(x_1) \psi(x_2) \psi(x_3) \psi(x_4)}_{\text{SCAMB}} A_\mu(z) \psi(z) \psi(z) A_\mu(w) \bar{\psi}(w) \psi(w)}_{\text{SCAMB}}}_{\text{SCAMB}}}_{\text{SCAMB}} \right] | 0 \rangle$$

$$\begin{cases} \psi(z) \bar{\psi}(x_1) & 5 & \text{SCAMB} \\ \psi(w) \bar{\psi}(x_2) & 5 & \text{SCAMB} \\ \psi(x_4) \bar{\psi}(z) & 0 & \text{SCAMB} \\ \psi(x_3) \bar{\psi}(w) & 0 & \text{SCAMB} \end{cases} \Rightarrow (-1)^{10} = +1$$

DOBBIAMO CALCOLARE

$$|\mathcal{M}|^2 = \frac{1}{4} \left\{ |\mathcal{M}_t|^2 + |\mathcal{M}_\mu|^2 - 2 \operatorname{Re} \mathcal{M}_t^\dagger \mathcal{M}_\mu \right\}$$

con

$$\mathcal{M}_t = \frac{i(-ie)^2}{(p_1 - p_3)^2 + i\varepsilon} (\bar{u}(p_3) \gamma^\mu u(p_1)) (\bar{u}(p_4) \gamma_\mu u(p_2))$$

$$\mathcal{M}_\mu = \frac{i(-ie)^2}{(p_1 - p_4)^2 + i\varepsilon} (\bar{u}(p_4) \gamma^\mu u(p_1)) (\bar{u}(p_3) \gamma_\mu u(p_2))$$

COMINCIAMO CON:

$$\frac{1}{4} \sum |\mathcal{M}_t|^2 = \frac{(-ie)^4}{4t^2} \sum \left(\bar{u}_\alpha(p_3) (\gamma^\mu)_{\alpha\beta} u_\beta(p_1) \right) \left(\bar{u}_\delta(p_4) (\gamma_\mu)_{\delta\gamma} u_\gamma(p_2) \right) \times \\ \times \left(\bar{u}_\zeta(p_1) (\gamma^\nu)_{\zeta\sigma} u_\sigma(p_3) \right) \left(\bar{u}_\varepsilon(p_2) (\gamma_\nu)_{\varepsilon\tau} u_\tau(p_4) \right)$$

$$\sim \frac{e^4}{4t^2} (\not{p}_1)_{\beta\delta} (\gamma^\nu)_{\zeta\sigma} (\not{p}_3)_{\sigma\alpha} (\gamma^\mu)_{\alpha\beta} \cdot (\not{p}_2)_{\gamma\varepsilon} (\gamma_\nu)_{\varepsilon\tau} (\not{p}_4)_{\tau\delta} (\gamma_\mu)_{\delta\gamma}$$

$$= \frac{e^4}{4t^2} (p_1)_\delta (p_2)^\sigma (p_3)_\delta (p_4)^\tau \text{Tr} \{ \gamma^\delta \gamma^\nu \gamma^\delta \gamma^\mu \} \text{Tr} \{ \gamma_\sigma \gamma_\nu \gamma_\tau \gamma_\mu \}$$

$$= \frac{4e^4}{t^2} (p_1)_\delta (p_2)^\sigma (p_3)_\delta (p_4)^\tau \left[\eta^{\delta\nu} \eta^{\delta\mu} - \eta^{\delta\delta} \eta^{\nu\mu} + \eta^{\delta\mu} \eta^{\nu\delta} \right] \times \\ \times \left[\eta_{\sigma\nu} \eta_{\tau\mu} - \eta_{\sigma\tau} \eta_{\nu\mu} + \eta_{\sigma\mu} \eta_{\nu\tau} \right]$$

$$= \frac{4e^4}{t^2} (p_1)_\delta (p_2)^\sigma (p_3)_\delta (p_4)^\tau \left[2\eta^\delta_\sigma \eta^\delta_\tau + 2\eta^\delta_\sigma \eta^\delta_\tau \right]$$

$$= \frac{8e^4}{t^2} \left\{ (p_1 p_2)(p_3 p_4) + (p_1 p_4)(p_2 p_3) \right\}$$

$$= \frac{2e^4}{t^2} \left\{ s^2 + u^2 \right\}$$

ANALOGAMENTE

$$\frac{1}{4} \sum |\mathcal{M}_\mu|^2 = \frac{e^4}{4\mu^2} \sum \left(\bar{u}_\alpha(p_4) (\gamma^\mu)_{\alpha\beta} u_\beta(p_1) \right) \left(\bar{u}_\delta(p_3) (\gamma_\mu)_{\delta\sigma} u_\sigma(p_2) \right) \times \\ \times \left(\bar{u}_\varepsilon(p_1) (\gamma^\nu)_{\varepsilon\tau} u_\tau(p_4) \right) \left(\bar{u}_\lambda(p_2) (\gamma_\nu)_{\lambda\delta} u_\delta(p_3) \right)$$

$$\sim \frac{e^4}{4\mu^2} (\not{p}_1)_{\beta\varepsilon} (\gamma^\nu)_{\varepsilon\tau} (\not{p}_4)_{\tau\alpha} (\gamma^\mu)_{\alpha\beta} \cdot (\not{p}_2)_{\sigma\lambda} (\gamma_\nu)_{\lambda\delta} (\not{p}_3)_{\delta\varrho} (\gamma_\mu)_{\varrho\sigma}$$

$$= \frac{e^4}{4\mu^2} (p_1)_\delta (p_4)_\sigma (p_2)^\varepsilon (p_3)^\tau \text{Tr} \{ \gamma^\delta \gamma^\nu \gamma^\sigma \gamma^\mu \} \text{Tr} \{ \gamma_\varepsilon \gamma_\nu \gamma_\tau \gamma_\mu \}$$

$$= \frac{4e^4}{\mu^2} (p_1)_\delta (p_4)_\sigma (p_2)^\varepsilon (p_3)^\tau \left[\eta^{\delta\nu} \eta^{\sigma\mu} - \eta^{\delta\sigma} \eta^{\nu\mu} + \eta^{\delta\mu} \eta^{\nu\sigma} \right] \left[\eta_{\varepsilon\nu} \eta_{\tau\mu} - \eta_{\varepsilon\tau} \eta_{\nu\mu} + \eta_{\varepsilon\mu} \eta_{\nu\tau} \right]$$

$$= \frac{4e^4}{u^2} (p_1)_\rho (p_4)_\sigma (p_2)^\varepsilon (p_3)^\tau \left[2\eta^\rho{}_\varepsilon \eta^\sigma{}_\tau + 2\eta^\rho{}_\tau \eta^\sigma{}_\varepsilon \right]$$

$$= \frac{8e^4}{u^2} \left[(p_1 p_2)(p_3 p_4) + (p_1 p_3)(p_2 p_4) \right]$$

$$= \frac{2e^4}{u^2} (s^2 + t^2)$$

E IL TERMINE MISTO

$$\frac{1}{4} \sum M_t^\dagger M_u = \frac{e^4}{4ut} \sum \left(\bar{u}_\alpha(p_1) (\gamma^\mu)_{\alpha\beta} u_\beta(p_3) \right) \left(\bar{u}_\delta(p_2) (\gamma_\mu)_{\delta\sigma} u_\sigma(p_4) \right) \times \\ \times \left(\bar{u}_\varepsilon(p_4) (\gamma^\nu)_{\varepsilon\tau} u_\tau(p_1) \right) \left(\bar{u}_\rho(p_3) (\gamma_\nu)_{\rho\gamma} u_\gamma(p_2) \right)$$

$$\sim \frac{e^4}{4ut} (p_3)_\beta (p_2)_\sigma (p_4)_\varepsilon (p_1)_\tau (\gamma^\mu)_{\alpha\beta} (\gamma_\mu)_{\delta\sigma} (\gamma_\nu)_{\varepsilon\tau} (\gamma^\nu)_{\rho\gamma}$$

$$= \frac{e^4}{4ut} (p_3)_\beta (p_2)_\sigma (p_4)_\varepsilon (p_1)_\tau \text{Tr} \{ \gamma_\beta \gamma_\sigma \gamma_\varepsilon \gamma_\tau \gamma^\mu \gamma^\nu \gamma_\nu \gamma_\sigma \}$$

$$= \frac{e^4}{4\mu t} (p_1)_\tau (p_2)_\sigma (p_3)_\delta (p_4)_\varepsilon (-32 \eta^{\sigma\tau} \eta^{\delta\varepsilon})$$

$$= -\frac{8e^4}{\mu t} (p_1 p_2) (p_3 p_4)$$

$$= -\frac{2e^4}{\mu t} s^2$$

DUNQUE ABBIAMO

$$|M|^2 = 2e^4 \left\{ \frac{s^2 + \mu^2}{t^2} + \frac{s^2 + t^2}{\mu^2} + \frac{2s^2}{\mu t} \right\}$$

$$= 2e^4 \left\{ \frac{s^2 \mu^2 + \mu^4 + s^2 t^2 + t^4 + 2s^2 \mu t}{(\mu t)^2} \right\}$$

$$= 2e^4 \frac{s^2(\mu^2 + t^2 + 2\mu t) + \mu^4 + t^4}{(\mu t)^2}$$

$$= 2e^4 \frac{\mu^4 + t^4 + s^2(\mu + t)^2}{\mu^2 t^2}$$

$$= 2e^4 \frac{\mu^4 + t^4 + s^4}{\mu^2 t^2}$$

RISULTATO SREDNICKI

$$|M|^2 = ze^4 \frac{s^4 + t^4 + u^4}{t^2 u^2}$$



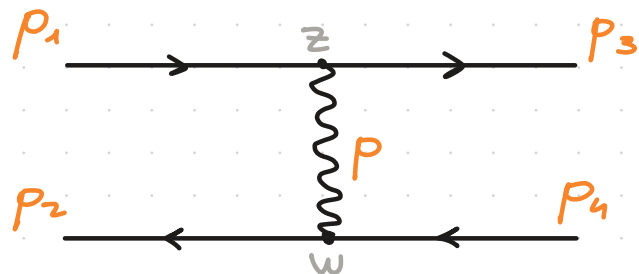
LA SEZIONE D'URTO È

$$\left. \frac{d\sigma}{d\Omega} \right|_{cm} = \frac{1}{64\pi s} ze^4 \frac{s^4 + t^4 + u^4}{t^2 u^2}$$

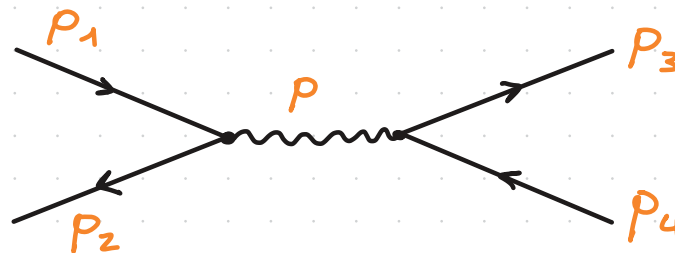
BHABHA SCATTERING

$$e^+ e^- \longrightarrow e^+ e^-$$

I DIAGRAMMI SONO :



CANALE t : $p = p_1 - p_3$



CANALE s : $p = p_1 + p_2$

CONTROLLIAMO IL SEGNO DEI DIAGRAMMI :

$$t : \quad \langle 0 | T \left[\underbrace{\psi(x_1) \psi(x_2) \psi(x_3) \psi(x_4) A_\mu(z) \psi(z) \psi(z) A_\nu(w) \psi(w) \psi(w)} \right] | 0 \rangle$$

$\psi(z) \psi(x_1)$	5 SCAMBI
$\psi(x_2) \psi(w)$	3 SCAMBI
$\psi(x_3) \psi(z)$	1 SCAMB.
$\psi(w) \psi(x_4)$	0 SCAMBI

$$\Rightarrow (-1)^9 = -1$$

$$s: \quad \langle 0 | T \left[\overline{\psi}(x_1) \psi(x_2) \overline{\psi}(x_3) \overline{\psi}(x_4) A_\mu(z) \overline{\psi}(z) \psi(z) A_\nu(w) \overline{\psi}(w) \psi(w) \right] | 0 \rangle$$

$$\begin{cases} \overline{\psi}(z) \psi(x_1) & 5 \text{ SCAMBI} \\ \overline{\psi}(x_2) \overline{\psi}(z) & 2 \text{ SCAMBI} \\ \overline{\psi}(x_3) \overline{\psi}(w) & 1 \text{ SCAMBI} \\ \overline{\psi}(w) \psi(x_4) & 0 \text{ SCAMBI} \end{cases} \Rightarrow (-1)^8 = +1$$

DUNQUE C'È UN SEGNO RELATIVO E DOBBIAMO CALCOLARE:

$$|\mathcal{M}|^2 = \frac{1}{4} \sum_s \left\{ |\mathcal{M}_s|^2 + |\mathcal{M}_t|^2 - 2 \operatorname{Re} \mathcal{M}_s^\dagger \mathcal{M}_t \right\}$$

con

$$\mathcal{M}_s = \frac{-i (-ie)^2}{(p_1 + p_2)^2 + i\varepsilon} \left(\bar{v}_\alpha(p_2) \gamma_{\alpha\beta}^\mu u_\beta(p_1) \right) \eta_{\mu\nu} \left(\bar{u}_\delta(p_3) \gamma_{\delta\sigma}^\nu v_\sigma(p_4) \right)$$

$$\mathcal{M}_t = \frac{-i (-ie)^2}{(p_1 - p_3)^2 + i\varepsilon} \left(\bar{u}_\alpha(p_3) \gamma_{\alpha\beta}^\mu u_\beta(p_1) \right) \eta_{\mu\nu} \left(\bar{v}_\delta(p_2) \gamma_{\delta\sigma}^\nu v_\sigma(p_4) \right)$$

COMINCIAMO A CALCOLARE:

$$\begin{aligned}
\frac{1}{4} \sum |\mathcal{M}_s|^2 &= \frac{e^4}{4s^2} \sum (\bar{v}_\alpha(p_2) \gamma_{\alpha\beta}^\mu \mu_\beta(p_1)) (\bar{u}_\delta(p_3) (\gamma_\mu)_{\delta\sigma} v_\sigma(p_4)) \times \\
&\quad \times (\bar{u}_\lambda(p_1) \gamma_{\lambda\tau}^\nu v_\tau(p_2)) (\bar{v}_\varepsilon(p_4) (\gamma_\nu)_{\varepsilon\varrho} \mu_\varrho(p_3)) \\
&\sim \frac{e^4}{4s^2} (\not{p}_2)_{\tau\alpha} \gamma_{\alpha\beta}^\mu (\not{p}_1)_{\beta\lambda} \gamma_{\lambda\tau}^\nu (\not{p}_4)_{\sigma\varepsilon} (\gamma_\nu)_{\varepsilon\varrho} (\not{p}_3)_{\varrho\delta} (\gamma_\mu)_{\delta\sigma} \\
&= \frac{e^4}{4s^2} p_{2\varrho} p_{1\sigma} \text{Tr}(\gamma^\varrho \gamma^\mu \gamma^\sigma \gamma^\nu) p_4^\varepsilon p_3^\tau \text{Tr}(\gamma_\varepsilon \gamma_\nu \gamma_\tau \gamma_\mu) \\
&= \frac{4e^4}{s^2} p_{1\sigma} p_{2\varrho} p_3^\tau p_4^\varepsilon [\eta^{\varrho\mu} \eta^{\sigma\nu} - \eta^{\varrho\sigma} \eta^{\mu\nu} + \eta^{\varrho\nu} \eta^{\mu\sigma}] \times \\
&\quad \times [\eta_{\varepsilon\nu} \eta_{\tau\mu} - \eta_{\varepsilon\tau} \eta_{\nu\mu} + \eta_{\varepsilon\mu} \eta_{\nu\tau}] \\
&= \frac{4e^4}{s^2} p_{1\sigma} p_{2\varrho} p_3^\tau p_4^\varepsilon [2 \eta^\varrho_\tau \eta^\sigma_\varepsilon - 4 \eta_{\varepsilon\tau} \eta^{\varrho\sigma} + 2 \eta^\varrho_\varepsilon \eta^\sigma_\tau + \\
&\quad + \eta^{\mu\nu} \eta_{\mu\nu} \eta_{\varepsilon\tau} \eta^{\varrho\sigma}]
\end{aligned}$$

$$= \frac{4e^4}{s^2} 2 \left[(p_2 p_3)(p_1 p_4) + (p_2 p_4)(p_1 p_3) \right]$$

$$= \frac{8e^4}{s^2} \left[\frac{\mu^2}{4} + \frac{t^2}{4} \right]$$

$$= 2e^4 \frac{\mu^2 + t^2}{s^2}$$

PROCEDIAMO .

$$\frac{1}{4} \sum |\mathcal{M}_t|^2 = \frac{e^4}{4t^2} \sum \left(\bar{u}_\alpha(p_3) \gamma_{\alpha\beta}^\mu u_\beta(p_1) \right) \left(\bar{v}_\delta(p_2) (\gamma_\mu)_{\delta\gamma} v_\gamma(p_4) \right) \times$$

$$\times \left(\bar{u}_\sigma(p_1) \gamma_{\sigma\tau}^\nu u_\tau(p_3) \right) \left(\bar{v}_\varepsilon(p_4) (\gamma_\nu)_{\varepsilon\tau} v_\tau(p_2) \right)$$

$$\sim \frac{e^4}{4t^2} (\not{p}_1)_{\beta\delta} \gamma_{\sigma\tau}^\nu (\not{p}_3)_{\sigma\alpha} \gamma_{\alpha\beta}^\mu \cdot (\not{p}_4)_{\gamma\varepsilon} (\gamma_\nu)_{\varepsilon\tau} (\not{p}_2)_{\tau\delta} (\gamma_\mu)_{\delta\gamma}$$

$$= \frac{e^4}{4t^2} p_{1\beta} p_{3\sigma} \text{Tr}\{\gamma^\beta \gamma^\nu \gamma^\sigma \gamma^\mu\} p_4^\tau p_2^\varepsilon \text{Tr}\{\gamma_\tau \gamma_\nu \gamma_\varepsilon \gamma_\mu\}$$

$$= \frac{4e^4}{t^2} p_{1\beta} p_2^\varepsilon p_{3\sigma} p_4^\tau [\eta^{\beta\nu} \eta^{\sigma\mu} - \eta^{\beta\sigma} \eta^{\nu\mu} + \eta^{\beta\mu} \eta^{\nu\sigma}] \times$$

$$\times [\eta_{\tau\nu} \eta_{\varepsilon\mu} - \eta_{\tau\varepsilon} \eta_{\nu\mu} + \eta_{\tau\mu} \eta_{\nu\varepsilon}]$$

$$= \frac{4e^4}{t^2} p_{1\beta} p_2^\varepsilon p_{3\sigma} p_4^\tau [2\eta^\beta{}_\tau \eta^\sigma{}_\varepsilon - 4\eta^{\beta\sigma} \eta_{\varepsilon\tau} + 2\eta^\beta{}_\varepsilon \eta^\sigma{}_\tau + \eta^{\mu\nu} \eta_{\mu\nu} \eta^{\beta\sigma} \eta_{\varepsilon\tau}]$$

$$= \frac{8e^4}{t^4} [(p_1 p_4)(p_2 p_3) + (p_1 p_2)(p_3 p_4)]$$

$$= \frac{8e^4}{t^4} \left[\frac{s^2}{4} + \frac{s^2}{4} \right]$$

$$= 2e^4 \frac{s^2 + s^2}{t^2}$$

L'ULTIMO TERMINE :

$$\frac{1}{4} \sum \mathcal{M}_s^\dagger \mathcal{M}_t = \frac{+e^4}{4st} \sum \left(\bar{u}_\alpha(p_1) \gamma_{\alpha\beta}^\mu v_\beta(p_2) \right) \left(\bar{v}_\sigma(p_4) (\gamma_\mu)_{\sigma\delta} u_\delta(p_3) \right) \times$$

$$\times \left(\bar{u}_\delta(p_3) \gamma_{\delta\lambda}^\nu u_\lambda(p_1) \right) \left(\bar{v}_\varepsilon(p_2) (\gamma_\nu)_{\varepsilon\tau} v_\tau(p_4) \right)$$

$$\sim \frac{+e^4}{4st} (\not{p}_2)_{\beta\varepsilon} (\gamma_\nu)_{\varepsilon\tau} (\not{p}_4)_{\tau\sigma} (\gamma_\mu)_{\sigma\delta} (\not{p}_3)_{\delta\lambda} (\gamma^\nu)_{\lambda\alpha} (\not{p}_1)_{\alpha\beta} (\gamma^\mu)_{\alpha\beta}$$

$$= \frac{+e^4}{4st} p_2^\delta p_4^\sigma p_3^\varepsilon p_1^\tau \underbrace{\text{Tr} \{ \gamma_\delta \gamma^\nu \gamma_\sigma \gamma^\mu \gamma_\varepsilon \gamma_\nu \gamma_\tau \gamma_\mu \} }_{= -32 \eta_{\sigma\tau} \eta_{\varepsilon\delta}}$$

$$= \frac{+e^4}{4st} (-32) (p_1 p_4) (p_2 p_3)$$

$$= -2e^4 \frac{s}{st}$$

DUNQUE ABBIAMO

$$\begin{aligned} |M|^2 &= ze^4 \left\{ \frac{u^2 + t^2}{s^2} + \frac{s^2 + u^2}{t^2} + \frac{u^2}{st} \right\} \\ &= ze^4 \left\{ \frac{u^2 t^2 + t^4 + s^4 + s^2 u^2 + u^2 st}{s^2 t^2} \right\} \\ &= ze^4 \left\{ \frac{s^4 + t^4 + u^2 (s^2 + t^2 + st)}{s^2 t^2} \right\} \\ &= ze^4 \left\{ \frac{s^4 + t^4 + u^2 (s + t)^2}{s^2 t^2} \right\} \\ &= ze^4 \left\{ \frac{s^4 + t^4 + u^4}{s^2 t^2} \right\} \end{aligned}$$

RISULTATO SREDNICKI:

$$|M|^2 \sim ze^4 \left(\frac{u^4 + s^4 + t^4}{s^2 t^2} \right)$$



LA SEZIONE D'URTO DIFFERENZIALE

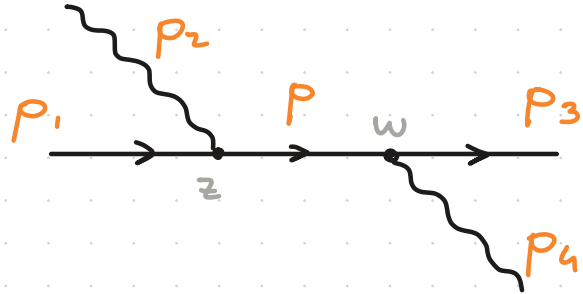
$$\left. \frac{d\sigma}{d\Omega} \right|_{cm} = \frac{ze^4}{64\pi s} \left\{ \frac{u^4 + s^4 + t^4}{s^2 t^2} \right\}$$

COMPTON SCATTERING

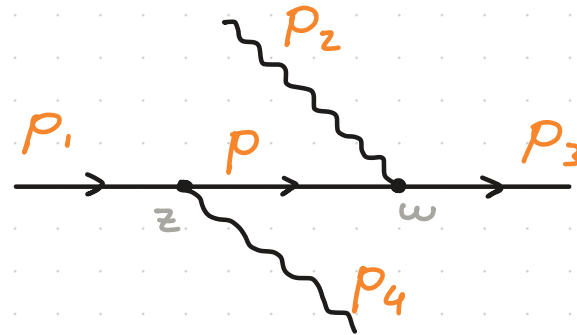
$$e^- \gamma \longrightarrow e^- \gamma$$

(FUNZIONA UGUALE
PER $e^+ \gamma \rightarrow e^+ \gamma$)

I DIAGRAMMI SONO :



$$\text{CANALE } s: p = p_1 + p_2$$



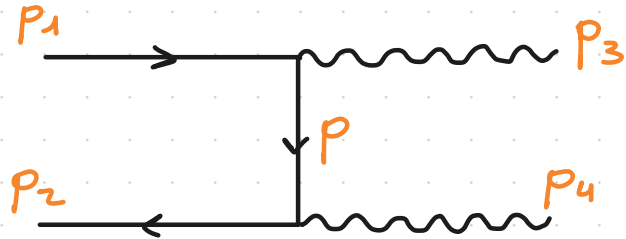
$$\text{CANALE } u: p = p_1 - p_4$$

CONTROLLIAMO IL SEGNO DEI DIAGRAMMI :

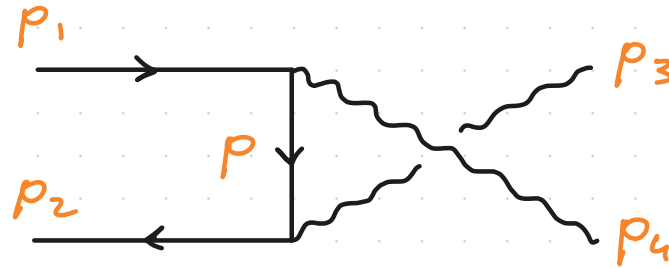
(NO NAME) SCATTERING

$$e^+ e^- \longrightarrow 2\gamma$$

I DIAGRAMMI SONO :



$$\text{CANALE } t: p = p_1 - p_3$$



$$\text{CANALE } u: p = p_1 - p_4$$

CONTROLLIAMO IL SEGNO DEI DIAGRAMMI :